

CSE 320 Quiz 2

Group 15

Full marks - 15

Time - 15 min

ID	
Name	

1	Consider a communication channel with a bandwidth of 8 kHz. The signal-to-noise ratio (SNR) of the channel is given as 25 dB. Calculate the maximum achievable bit rate using the Shannon capacity formula. However, due to the presence of interference, the actual SNR drops to 20 dB. How does this affect the maximum achievable bit rate?	5
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2	Given a signal in the time domain, express it in the frequency domain: $x(t) = 3\sin(40t + 3) + 4\cos(100t - 5)$	5
3	Digital audio is sampled at a rate of 44,100 samples per second. What is the range of frequencies that can be captured?	5